

注释 16.1

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说明 1. 证明：平面三角形不等式，设

$$P_1 = (x_1, y_1)$$

$$P_2 = (x_2, y_2)$$

$$P_3 = (x_3, y_3)$$

可得

$$\rho(P_1, P_2) \leq \rho(P_1, P_3) + \rho(P_2, P_3)$$

证明：我们有

$$\begin{aligned} & \rho(P_1, P_2) \\ &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(x_2 - x_3 + x_3 - x_1)^2 + (y_2 - y_3 + y_3 - y_1)^2} \\ &= \sqrt{(x_2 - x_3)^2 + (x_3 - x_1)^2 + 2(x_2 - x_3)(x_3 - x_1) + (y_2 - y_3)^2 + (y_3 - y_1)^2 + 2(y_2 - y_3)(y_3 - y_1)} \end{aligned}$$

和

$$\begin{aligned} \rho(P_1, P_3) &= \sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2} \\ \rho(P_2, P_3) &= \sqrt{(x_2 - x_3)^2 + (y_2 - y_3)^2} \end{aligned}$$

于是

$$\begin{aligned}
& [\rho(P_1, P_3) + \rho(P_2, P_3)]^2 - \rho^2(P_1, P_2) \\
&= 2\sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2}\sqrt{(x_2 - x_3)^2 + (y_2 - y_3)^2} - 2(x_2 - x_3)(x_3 - x_1) - 2(y_2 - y_3)(y_3 - y_1) \\
&= 2\sqrt{[(x_3 - x_1)^2 + (y_3 - y_1)^2][(x_2 - x_3)^2 + (y_2 - y_3)^2]} - 2(x_2 - x_3)(x_3 - x_1) - 2(y_2 - y_3)(y_3 - y_1)
\end{aligned}$$

由柯西-施瓦茨不等式（证明 base.tex 中有证明）

$$[(x_3 - x_1)^2 + (y_3 - y_1)^2][(x_2 - x_3)^2 + (y_2 - y_3)^2] \geq [(x_3 - x_1)(x_2 - x_3) + (y_3 - y_1)(y_2 - y_3)]^2$$

代入上式可得

$$[\rho(P_1, P_3) + \rho(P_2, P_3)]^2 - \rho^2(P_1, P_2) \geq 0$$

命题得证。